CSCI UA.0060

Assignment 5 - SQL Queries

Deliverables

Upload the following files to your GitHub repository for this assignment:

- A single **.sql** file containing all of your queries.
- A single **.txt** file containing (1) the output produced by each query and (2) a brief explanation, in your own words, of the SQL query logic.

Filename should be FirstInitialLastName-Assign5.sql. For example, John Smith's deliverables would be JSmith-Assign5.sql + JSmith-Assign5.txt.

Overall Requirements

In this assignment, you will develop SQL queries to respond to the questions listed below concerning the United Helpers activities. In order to ensure that there are clear, "correct" answers, you will use a standard database, not the one you created in the last assignment.

The .sql file to create the database can be forked from an EdStem repository here:

https://edstem.org/us/courses/105050/workspaces/pcb9R92a0kcUD9yh1DJKH8EB4ta7vpi8

Or, the .sql file is also available in this GitHub repository if you would prefer to work in your own development environment.

Copy the queries you develop into the .sql file. For each query, copy the returned output into the .txt file and immediately follow it with a brief explanation of what the query is doing. The explanation must rephrase the SQL logic in your own words; do not simply repeat the SQL statement.

> **Example explanation:** "This JOIN matches each volunteer to the tasks assigned to them by connecting rows with the same volunteer ID. It then displays the volunteer's name and the matching task description." Include the actual output from the query as well.

Upload both files to your GitHub repository. Upload to Brightspace the link to your GitHub repository.

Assignment Questions

1 How many containers of antibiotics are currently available?

2 Which volunteer(s), if any, have phone numbers that do not start with the number 2 and whose last name is not Jones. Query should retrieve names rather than ids.

3 Which volunteer(s) are working on transporting tasks? Query should retrieve names rather than ids.

4 Which task(s) have yet to be assigned to any volunteers (provide task descriptions, not the codes)?

5 Which type(s) of package contain some kind of bottle?

6 Which items, if any, are not in any packages? Answer should be item descriptions.

7 Which task(s) are assigned to volunteer(s) that live in New Jersey (NJ)? Answer should have the task description and not the task ids.

8 Which volunteers began their assignments in the first half of 2021? Answer should have the volunteer names and not their ids.

9 Which volunteers have been assigned to tasks that include packing spam? Answer should have the volunteer names and not their ids.

10 Which item(s) (if any) have a total value of exactly $100 in one package? Answer should be item descriptions.

11 How many volunteers are assigned to tasks with each different status? The answer should show each different status and the number of volunteers sorted from highest to lowest.

12 Which task creates the heaviest set of packages and what is the weight? Show both the taskCode and the weight (You should be able to do this without using any sub-queries).

13 How many tasks are there that do not have a type of "packing"?

14 Of those items that have been packed, which item (or items) were touched by fewer than 3 volunteers? Answer should be item descriptions.

15 Which packages have a total value of more than 100? Show the packageIds and their value sorted from lowest to highest.

Grading Rubric

See Brightspace for Grading Rubric